variables—instead, we effectively have infinitely many variables and can make our predictions *anywhere*, as illustrated in Figure 2.13. This is why while a GP has all the properties of a multivariate Gaussian, the GP exists in infinite dimensions.

For the same reason, a GP can be considered as a distribution over functions, as the title of this chapter suggests. The progression that we have gone through in this chapter, from a one-dimensional normal to a GP, is summarized in Table 2.1.

Table 2.1 Gaussian distribution objects and what they model. With a Gaussian process, we operate under infinite dimensions, modeling functions instead of numbers or vectors.

Distribution type	Number of modeled variables	Description
A one-dimensional normal distribution	One	A distribution over numbers
A multivariate Gaussian distribution	Finitely many	A distribution over vectors of finite length
A Gaussian process	Infinitely many	A distribution over functions

To see a GP in action, let's reexamine the curve-fitting procedure in Figure 2.1 at the beginning of this chapter, where we limit our domain to be between -5 and 5.

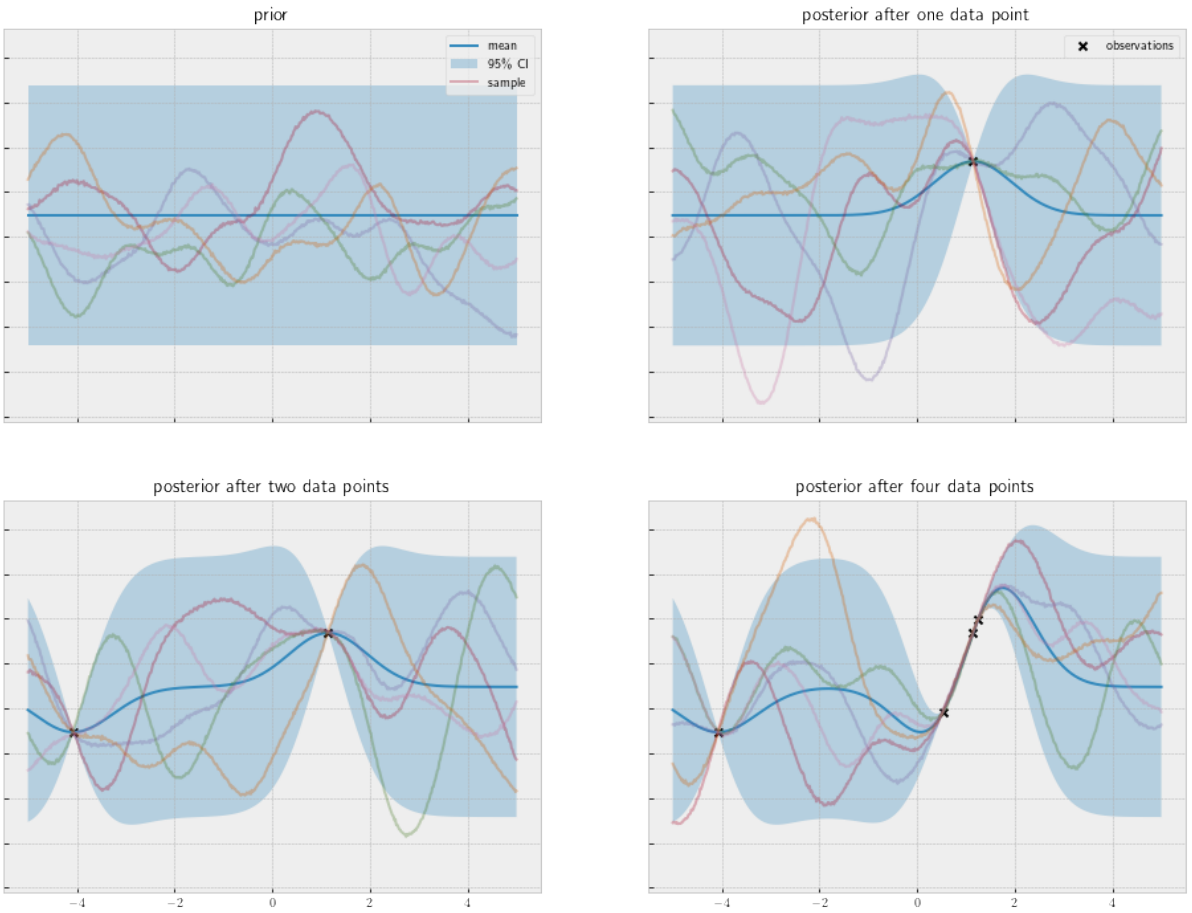


Figure 2.14 Predictions made by a GP conditioned on zero, one, two, and four observations.

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